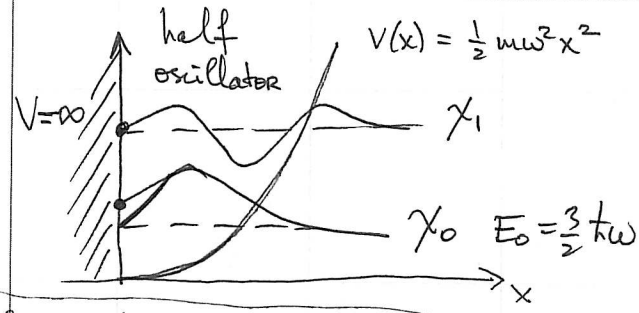
**P1**

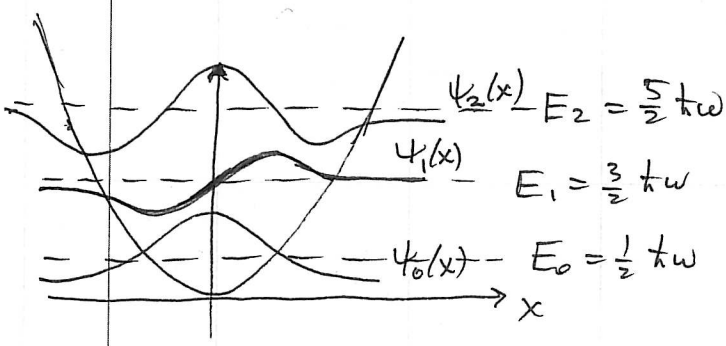
(a)




$$V(x) = \begin{cases} \frac{1}{2} m\omega^2 x^2 & , x>0 \\ \infty & , x<0 \end{cases}$$

$\boxed{\text{Half-oscillator is obtained from standard by imposing } \psi(0)=0}$ $\Rightarrow \chi(x=0)=0.$

Standard oscillator: even wave function (with $n=0,2,4\ldots$) have finite value at $x=0$; while odd ones (with $n=1,3,5\ldots$) $\psi_{odd}(x) = -\psi_{odd}(-x) \Rightarrow \psi_{odd}(x=0)=0$ turn to 0 at the origin.




$\Rightarrow$ for the half-oscillator $\chi_m(x) = c\,\psi_{2m+1}(x)$ $(n=2m+1)$

$\Rightarrow$ The lowest energy state $m=0$ $E_0 = \frac{3}{2}\hbar\omega$
The next level has $m=1$ $E_1 = \frac{5}{2}\hbar\omega$
$m=2$ $E_2 = \frac{7}{2}\hbar\omega$

$$E_m = \hbar\omega\left(2m+1+\tfrac{1}{2}\right) = \hbar\omega\left(2m+\tfrac{3}{2}\right)$$

(b) $\chi_m(x) = c\,\psi_{2m+1}(x)$

$|c|$ is found by requiring $\chi_m(x)$ be normalized:

$$1 = \int_0^\infty dx\,|\chi_m(x)|^2 = \int_0^\infty dx\,|c|^2\,|\psi_{2m+1}(x)|^2 = \frac{|c|^2}{2}\underbrace{\int_{-\infty}^{\infty} dx\,|\psi_{2m+1}(x)|^2}_{1}$$

$$= \frac{|c|^2}{2} \Rightarrow |c| = \sqrt{2}$$

$$\Rightarrow \chi_m(x) = \sqrt{2}\,\psi_{2m+1}(x)$$

P2 (a) $H_n(\xi) = (-1)^n e^{\xi^2} \left(\frac{d}{d\xi}\right)^n e^{-\xi^2}$

$$\Rightarrow H_2(\xi) = e^{\xi^2}\left(\frac{d}{d\xi}\right)^2 e^{-\xi^2} = e^{\xi^2}\frac{d}{d\xi}\left(-2\xi e^{-\xi^2}\right) = e^{\xi^2}\left(4\xi^2 e^{-\xi^2} - 2e^{-\xi^2}\right)$$

$$= 4\xi^2 - 2.$$

$$H_3(\xi) = (-1)^3 e^{\xi^2}\left(\frac{d}{d\xi}\right)^3 e^{-\xi^2} = (-)e^{\xi^2}\frac{d}{d\xi}\left(4\xi^2-2\right)e^{-\xi^2} =$$

$$= (-)e^{\xi^2}\left(8\xi e^{-\xi^2} - 2\xi(4\xi^2-2)e^{-\xi^2}\right) =$$

$$= (-)\left[8\xi - 8\xi^3 + 4\xi\right] = 8\xi^3 - 12\xi$$

(b) $\underset{\substack{\downarrow \\ n+1}}{H_4}(\xi) = 2\xi H_3(\xi) - 2\cdot 3\cdot H_2(\xi) =$

$$= 2\xi\left(8\xi^3 - 12\xi\right) - 6\left(4\xi^2-2\right) = 16\xi^4 - 24\xi^2 - 24\xi^2 + 12$$

$$= 16\xi^4 - 48\xi^2 + 12$$

$$H_5(\xi) = 2\xi H_4(\xi) - 2\cdot 4 H_3(\xi) = 2\xi\left(16\xi^4 - 48\xi^2 + 12\right)$$

$$-8\left(8\xi^3 - 12\xi\right) = 32\xi^5 - 96\xi^3 + 24\xi - 64\xi^3 + 96\xi$$

$$= 32\xi^5 - 160\xi^3 + 120\xi$$

(c) Check $\frac{dH_n(\xi)}{d\xi} = 2n H_{n-1}(\xi)$

$$\frac{d}{d\xi}H_5 = 32\cdot 5\xi^4 - 160\cdot 3\xi^2 + 120 =$$

$$= 10\left[16\xi^4 - 48\xi^2 + 12\right] = 2\cdot 5\cdot H_4(\xi)$$

$$\frac{d}{d\xi}H_4(\xi) = 16\cdot 4\xi^3 - 48\cdot 2\xi =$$

$$= 8\left[8\xi^3 - 12\xi\right] = 2\cdot 4\cdot H_3(\xi)$$

(d) $$e^{-z^2+2z\xi} = \sum_{n=0}^{\infty} \frac{z^n}{n!} H_n(\xi) =$$

$$= H_0(\xi) + \frac{z}{1!} H_1(\xi) + \frac{z^2}{2!} H_2(\xi)$$

But $e^{-z^2+2z\xi} = e^{-z^2} e^{2z\xi} = \left(1 - z^2 + O(z^4)\right)\left(1 + 2z\xi + \frac{1}{2}(2z\xi)^2 + O(z^3)\right) =$ (we need to keep up to $z^2$ terms only) $=$

$$= 1 - z^2 + 2z\xi + \frac{1}{2} 4z^2\xi^2 + O(z^3) =$$

$$= 1 + z(2\xi) + \frac{z^2}{2}(4\xi^2 - 2)$$

$$= H_0 + zH_1 + \frac{z^2}{2} H_2$$

$$\Rightarrow \begin{cases} H_0(\xi) = 1 \\ H_1(\xi) = 2\xi \\ H_2(\xi) = 4\xi^2 - 2 \end{cases}$$

$\boxed{P3}$ We know initial $(t=0)$ state (wave function)

$$\psi_{init}(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega}{2\hbar}x^2}$$

When frequency doubled, $\omega \to \omega' = 2\omega$, the eigenstates of the system are given by those of the harmonic oscillator with frequency $\boxed{\omega' = 2\omega}$: we denote them $\psi_n'(x)$.

[They are given by Eq. (7.45) with replacement: $\omega \to \omega'$]

To find probabilities of various (energy) measurements, we have to expand $\psi_{init}(x)$ in terms of $\psi_n'(x)$

$$\psi_{init}(x) = \sum_{n=0}^{\infty} c_n \psi'_n(x)$$

Then needed probability is given by $\underline{P_n = |c_n|^2}$.

$\{\psi'_n(x)\}$ have following states:

$$\begin{cases} \psi'_0(x) & , E_0 = \hbar\omega'/2 = \frac{1}{2}\hbar\omega' = \hbar\omega \\ \psi'_1(x) & , E_1 = \frac{3}{2}\hbar\omega' = 3\hbar\omega \\ \psi'_2(x) & , E_2 = \frac{5}{2}\hbar\omega' = 5\hbar\omega \\ \vdots & \end{cases}$$

(a) Hence the probability to observe $E = \frac{1}{2}\hbar\omega = \frac{1}{4}\hbar\omega'$ is <u>ZERO</u> because there is no corresponding eigenstate $\psi'(x)$.

(b) Probability to get $\hbar\omega = \frac{1}{2}\hbar\omega'$ $\Rightarrow$ probability that $\psi_{init}$ "evolved" into $\psi'_0(x)$ [which corresponds to eigenvalue $\frac{1}{2}\hbar\omega'$].

$\Rightarrow$ we need $c_0 = \langle \psi'_0 | \psi_{init} \rangle =$

$$= \int_{-\infty}^{\infty} dx \left[\psi'_0(x)\right]^* \psi_{init}(x)$$

Clearly $\psi'_0(x) = \left(\frac{m\omega'}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega'}{2\hbar}x^2}$

It helps to introduce $\xi = \sqrt{\frac{m\omega'}{\hbar}}\, x$

Then $\psi_{init}(x) = \left(\frac{m\omega'}{2\pi\hbar}\right)^{1/4} e^{-\frac{m\omega'}{4\hbar}x^2} = \left(\frac{m\omega'}{2\pi\hbar}\right)^{1/4} e^{-\xi^2/4}$

$$\psi_0'(x) = \left(\frac{m\omega'}{\pi\hbar}\right)^{1/4} e^{-\frac{m\omega'}{2\hbar}x^2} = \left(\frac{m\omega'}{\pi\hbar}\right)^{1/4} e^{-\xi^2/2}$$

$$\Rightarrow c_0 = \left(\frac{m\omega'}{2\pi\hbar}\right)^{1/4}\left(\frac{m\omega'}{\pi\hbar}\right)^{1/4} \int_{-\infty}^{\infty} dx\, e^{-\xi^2/2}\, e^{-\xi^2/4} =$$

$$= \left(x = \sqrt{\frac{\hbar}{m\omega'}}\,\xi\right) = \left(\frac{m\omega'}{\pi\hbar}\right)^{1/2} \frac{1}{2^{1/4}} \sqrt{\frac{\hbar}{m\omega'}} \int_{-\infty}^{\infty} d\xi\, e^{-\frac{3}{4}\xi^2} =$$

$$= \frac{1}{2^{1/4}\pi^{1/2}} \underbrace{\int_{-\infty}^{\infty} d\xi\, e^{-\frac{3}{4}\xi^2}}_{\sqrt{\frac{4\pi}{3}}} = \frac{1}{2^{1/4}} \frac{2}{\sqrt{3}} = \frac{2^{3/4}}{3^{1/2}}$$

$$\Rightarrow P_0\left(E = \hbar\omega'/2 = \hbar\omega\right) = |c_0|^2 = \frac{2^{3/2}}{3} = 0.943$$

(c) $P_1\left(E = \frac{3}{2}\hbar\omega' = 3\hbar\omega\right) = |c_1|^2$ ~~#~~

but $c_1 = \int_{-\infty}^{\infty} dx\, \psi_1'(x)\, \psi_{init}(x) = 0$ because

$\psi_1'(x) =$ <u>odd</u> function (corresponds to $E = \frac{3}{2}\hbar\omega'$)

while $\psi_{init}(x) =$ <u>even</u> $\Rightarrow P_1 = 0.$

Finally $P_2\left(E = \frac{5}{2}\hbar\omega' = 5\hbar\omega\right) = |c_2|^2$

$$c_2 = \int_{-\infty}^{\infty} dx\, \psi_2'(x)\, \psi_{init}(x)$$

From <u>Prob. 2</u> $\psi_2'(x) = \left(\frac{m\omega'}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^2 \cdot 2!}} H_2(\xi)\, e^{-\xi^2/2}$ where $\xi = \sqrt{\frac{m\omega'}{\hbar}}\, x$

$$H_2(\xi) = 4\xi^2 - 2$$

$$\Rightarrow \psi_2'(x) = \left(\frac{m\omega'}{\pi\hbar}\right)^{1/4} \frac{4\xi^2-2}{\sqrt{8}} e^{-\xi^2/2} = \left(\frac{m\omega'}{\pi\hbar}\right)^{1/4} \frac{2\xi^2-1}{\sqrt{2}} e^{-\xi^2/2}$$

$$\Rightarrow c_2 = \left(\frac{m\omega'}{\pi\hbar}\right)^{1/4} \cdot \frac{1}{\sqrt{2}} \left(\frac{m\omega'}{2\pi\hbar}\right)^{1/4} \int_{-\infty}^{\infty} dx \, (2\xi^2-1) e^{-\xi^2/2} \cdot e^{-\xi^2/4}$$

$$= \frac{1}{\sqrt{2}} \frac{1}{(2\pi^2)^{1/4}} \left(\frac{m\omega'}{\hbar}\right)^{1/2} \sqrt{\frac{\hbar}{m\omega'}} \int_{-\infty}^{\infty} d\xi \, (2\xi^2-1) e^{-\frac{3}{4}\xi^2} =$$

$$= \frac{1}{2^{3/4}\sqrt{\pi}} \int_{-\infty}^{\infty} d\xi \, (2\xi^2-1) e^{-\frac{3}{4}\xi^2}$$

$$I_1 = \int_{-\infty}^{\infty} d\xi \, e^{-\frac{3}{4}\xi^2} = \sqrt{\frac{4\pi}{3}}$$

$$I_2 = \int_{-\infty}^{\infty} d\xi \, 2\xi^2 e^{-\frac{3}{4}\xi^2} = -2 \frac{d}{da} \int_{-\infty}^{\infty} d\xi \, e^{-a\xi^2} \Big|_{a=3/4} =$$

$$= -2 \frac{d}{da} \sqrt{\frac{\pi}{a}} \Big|_{a=3/4} = -2\sqrt{\pi} \left(-\frac{1}{2a^{3/2}}\right)\Big|_{a=3/4} =$$

$$= \frac{\sqrt{\pi}}{\left(\frac{3}{4}\right)^{3/2}} = \left(\frac{4}{3}\right)^{3/2} \sqrt{\pi}$$

$$\Rightarrow c_2 = \frac{1}{2^{3/4}\sqrt{\pi}} \left(\left(\frac{4}{3}\right)^{3/2} \sqrt{\pi} - \sqrt{\frac{4\pi}{3}}\right) = \frac{1}{2^{3/4}} \left(\frac{4}{3}\right)^{1/2} \left[\frac{4}{3} - 1\right] =$$

$$= \frac{1}{2^{3/4}} \frac{2}{\sqrt{3}} \frac{1}{3} = \frac{2^{1/4}}{3^{3/2}}$$

$$\Rightarrow P_2 \left(E = \frac{5}{2}\hbar\omega' = 5\hbar\omega\right) = \left(\frac{2^{1/4}}{3^{3/2}}\right)^2 = \frac{2^{1/2}}{3^3} = \frac{\sqrt{2}}{3^3} = \frac{\sqrt{2}}{27}$$

$$P_2 = 0.052$$

Note that $P_0 + P_2 = 0.943 + 0.052 = 0.995 < 1$